

Mateusz Grzonka

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SUMMARY

Backend-focused Computer Science student at Gdańsk University of Technology. Interested in music theory, game development, automotive engineering and tools that actually improve everyday life.

EDUCATION

Gdańsk University of Technology | B.Eng. in Computer Science **2023 – Expected 2027**
• Thesis title: A 3D FDTD-based acoustic wave propagation simulator
ZSŁ Technical Highschool in Gdańsk | IT Technician **2019 – 2023**

PROJECTS

A 3D FDTD-based Acoustic Wave Propagation Simulator *Python, CUDA, Taichi Lang*
• A high-performance 3D simulator utilizing the FDTD method to model acoustic wave behavior in complex environments. Consists of a Voxelizer, Wave Propagation Engine and Visualizer. Currently working on a RL Agent that helps you adapt your room for better general acoustics.
Secure TTP/Server/Client Authentication Protocol *C#, .Net*
• Designed and implemented a custom, three-node network protocol (Client, Server, TTP) with secure, encrypted communication channels using hybrid cryptography (RSA-4096 & AES-256).
Multiplayer Game with UDP Server *Python, Godot*
• Godot game and custom Python backend server to handle real-time multiplayer networking utilizing UDP protocols with custom frames to manage game state synchronization.
MACD Market Trading Strategy Calculator *Python*
• Developed an algorithmic calculator utilizing the MACD method to analyze market data. Implemented features to generate and visualize buy/sell signals based on simulated trading scenarios.

EXPERIENCE

Coding Giants | *Math and Programming Teacher* **2023 – 2025**
• Translated complex computer science concepts into easily understandable lessons. *Python*
Intel | *Hardware Specialist Internship* **2021**
• Collaborated with the engineering team to troubleshoot hardware integration issues. *Hardware*

SKILLS

- **Languages:** Python, C/C++, C#, .Net, Java, Spring, Taichi Lang, HTML, CSS, JavaScript
- **Technologies & Tools:** Linux, Docker, Git, VirtualBox, Markdown
- **Soft Skills & Misc:** Working in a group, Teaching, Research, Technical Documentation
- **Spoken Languages:** English (C1), Polish (Native)

EVENTS

- Took part in hackathon "Hack4Change 2025" organized by CODE:ME.
- Took part in three different "Infoshare" Tech Events.